

Document History

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Overview

This document summarizes the testing and test results that were conducted as the ASL AZFP module 'GLIDER-12' was integrated into a Slocum Glider. Testing was carried out at Teledyne Marine facilities by Teledyne personnel, with remote support from ASL personnel. Data analysis was performed by ASL.

Executive Summary

- Initial testing of the GLIDER-12 module at Teledyne Marine was performed after the module had been integrated into the payload bay of the Slocum. Results were consistent with expectations; basic transmit functionality was verified, and noise floor measurements were roughly consistent with results obtained during calibration at ASL.
- System-level testing occurred next, after the module had been fully integrated into the Slocum glider. Basic transmit functionality was verified, and noise floor measurements were roughly consistent with results obtained during calibration at ASL.
- Interference was observed during system-level testing but is likely manageable, believed to originate primarily from the 172 kHz altimeter onboard the glider.

Background

ASL and Teledyne Marine are collaborating to introduce and to improve performance testing on AZFP modules that are integrated into Slocum gliders. The AZFP module 'GLIDER-12' is the first such module to undergo this new and evolving series of acoustic performance tests, which are performed at Teledyne Marine facilities as part of the integration process. Noise level measurements are then compared against noise level measurements that occurred during calibration of the module at ASL's facilities.

Test 1: GLIDER-12 Module Integrated into Payload Bay

The initial test was conducted with the AZFP module integrated into the payload bay, running off battery power, and with no RS-232 connection. All other payload bay sensors were 'OFF' at the time of these tests. The purpose of these tests was to perform an 'incoming inspection' of sorts; to verify that the AZFP module has not obviously been damaged in shipping, and to verify that it is functioning in a manner consistent with expectation following its evaluation at ASL.

The test consisted of a single simple 'deployment' of the AZFP, with each of the three channels transmitting a pulse of non-zero length (to verify that each transmitter is operating) and receiving out to 500 m range (to assess the noise floor for each channel). The payload bay was sealed and was deployed into a small tank of water.

The outcome of the test was positive; the AZFP does not appear to have been damaged in shipping and was observed to operate in a manner that is consistent with expectation. In particular, each channel's transmitter saturates the A/D at the start of each ping, the signal for each channel is observed to reach the noise floor within approximately 80 m, and the noise floor for each channel is consistent with expectations.

During calibration at ASL, the noise floor for each channel was measured. The 67 kHz channel had the lowest noise floor, at approximately 10,000 A/D counts. The 38 kHz channel had a noise floor of approximately 12,000 A/D counts. The 125 kHz channel had a noise floor of approximately 14,000 A/D counts, although occasional 'spikes' in noise level were observed up to approximately 17,000 counts.

When compared against noise levels measured during Test 1 (AZFP module integrated into payload bay) noise levels were found to be consistent with noise levels measured at ASL. No obvious interference was observed. Screenshots from this test are attached below.

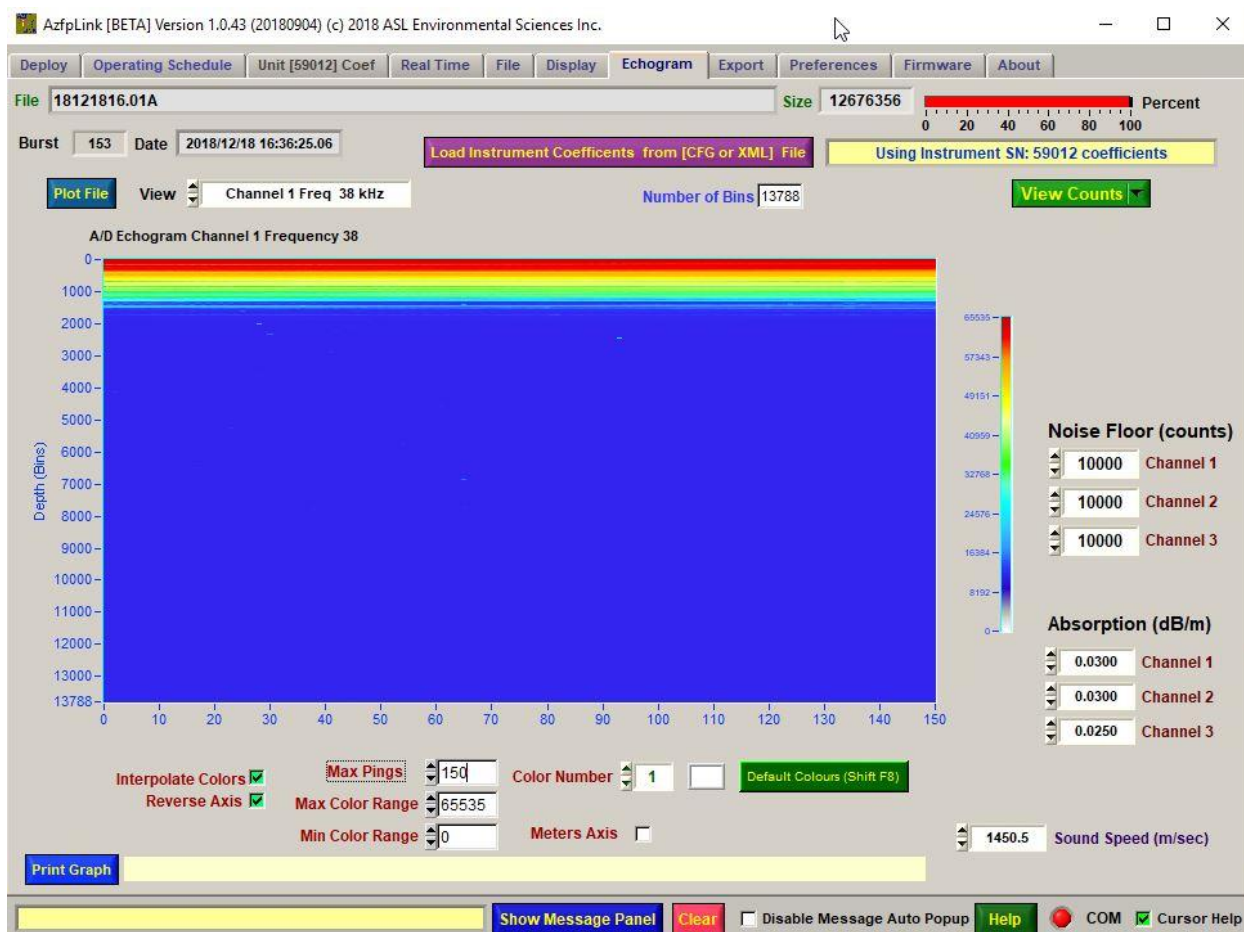


Figure 1: Echogram from Test 1 for the 38 kHz channel.

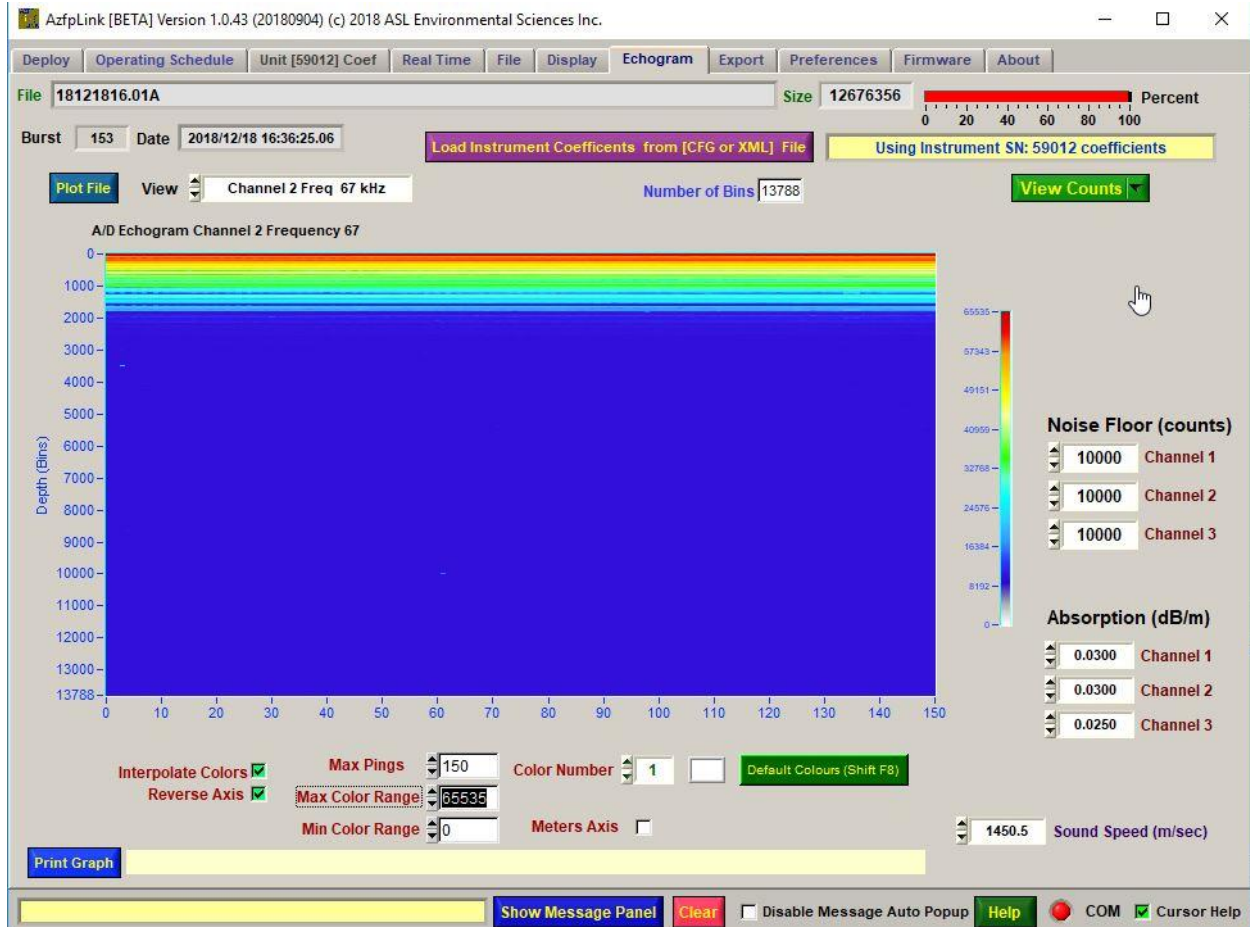


Figure 2: Echogram from Test 1 for the 67 kHz channel.

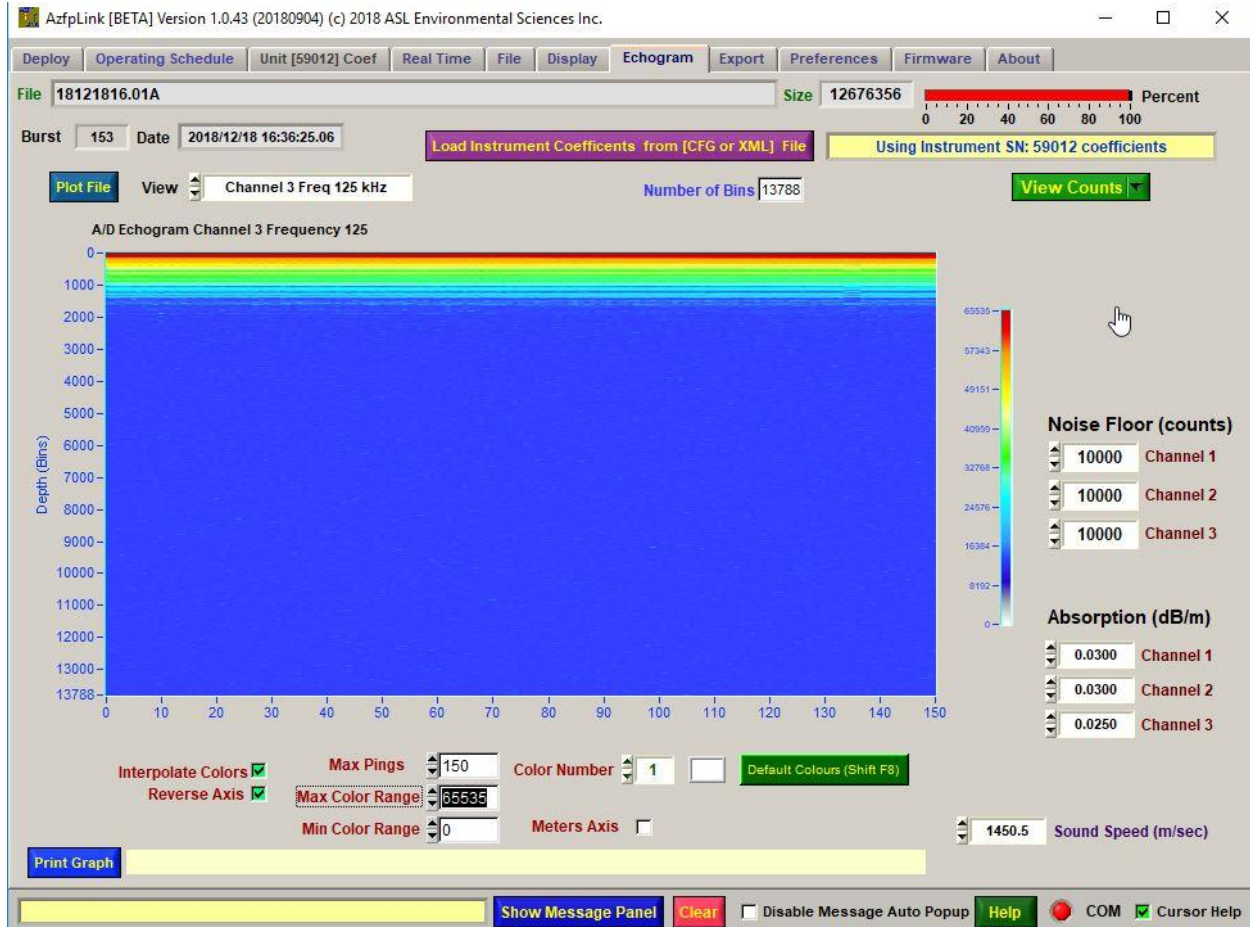


Figure 3: Echogram from Test 1 for the 125 kHz channel.

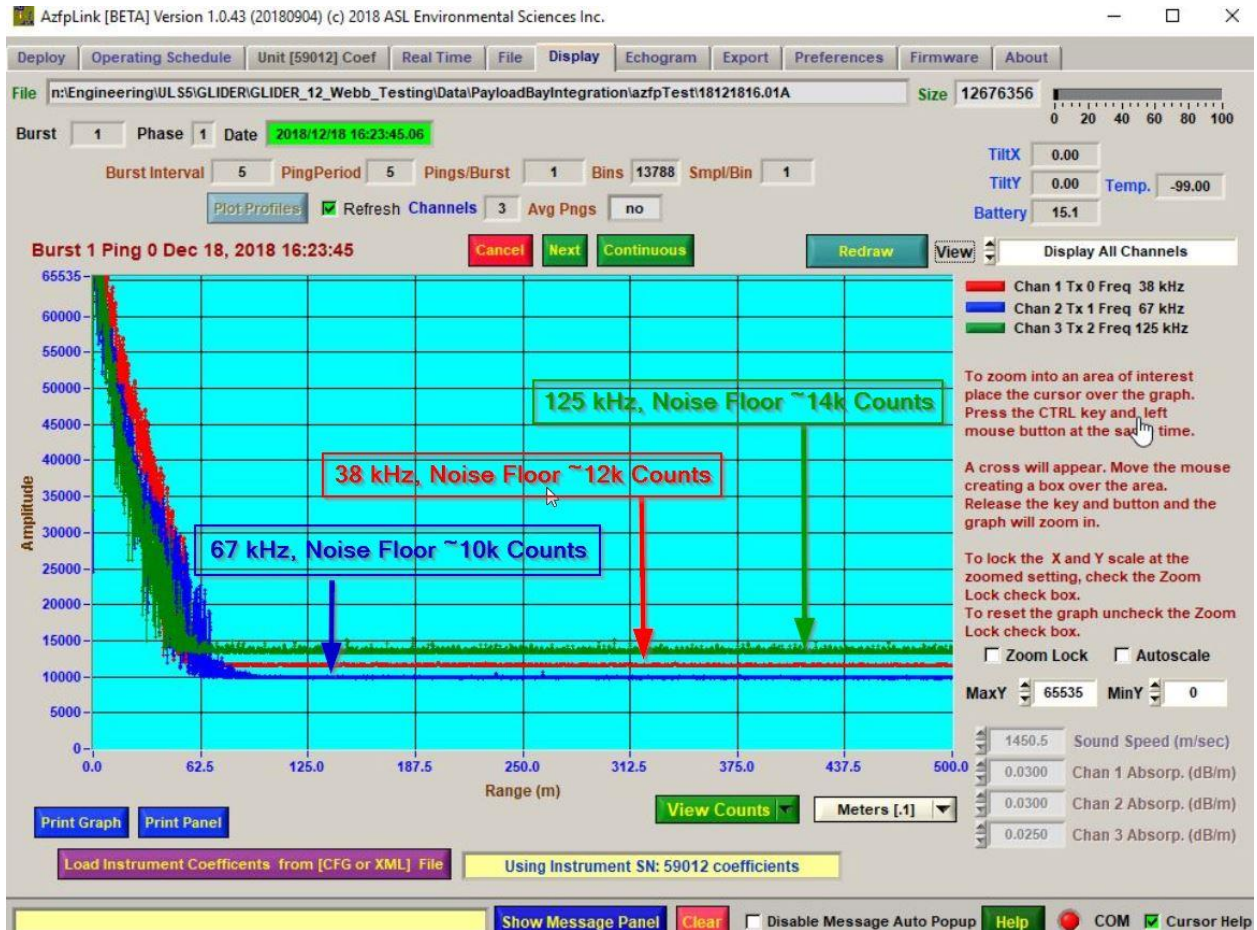


Figure 4: Annotated profile for each channel (vertical axis in units of A/D counts) during Test 1.

Test 2: Payload Bay Integrated into Glider

These tests of the GLIDER-12 AZFP module were conducted with the glider fully assembled and deployed in TWR's 5 x 5 x 5 m tank. Results from two separate tests of the fully-assembled glider are presented.

Test 2A: Tx Test

The first such test occurred with the AZFP powered by the glider and with other sensors (e.g. 172 kHz altimeter) 'ON', but without pumps or thrusters enabled and without adjusting battery position. The glider was initially out of the water during this first test, and was lowered into the water about 20% of the way through the AZFP's deployment. This deployment resembled the Payload Bay test (Test 1, above); the transmitter was again enabled and the AZFP was collecting data to 500 m range on each channel.

Results indicate that the AZFP was running as expected post-integration into the glider. Noise floors for each channel are consistent with expectation and with prior testing of the GLIDER-12 module. The transmitted pulse is seen to reverberate in the tank with 'spikes' approximately every 6 m. This range is roughly consistent with tank dimensions. The strength of the signal from the transmitted pulse is observed to decay to the noise floor within approximately 125 m on each channel.

Noise floors for each channel are assessed following the decay of the reverberation. Each channel exhibits a noise floor that is consistent with prior testing of the GLIDER-12 module (10k / 12k / 14k counts on the 67 / 38 / 125 kHz channels, respectively), but transient interference is observed on each channel. This interference takes the form of two or three (channel-dependent) unique increases in the amplitude of the Rx signal. The source of this interference is presently unknown. See below for screen shots.

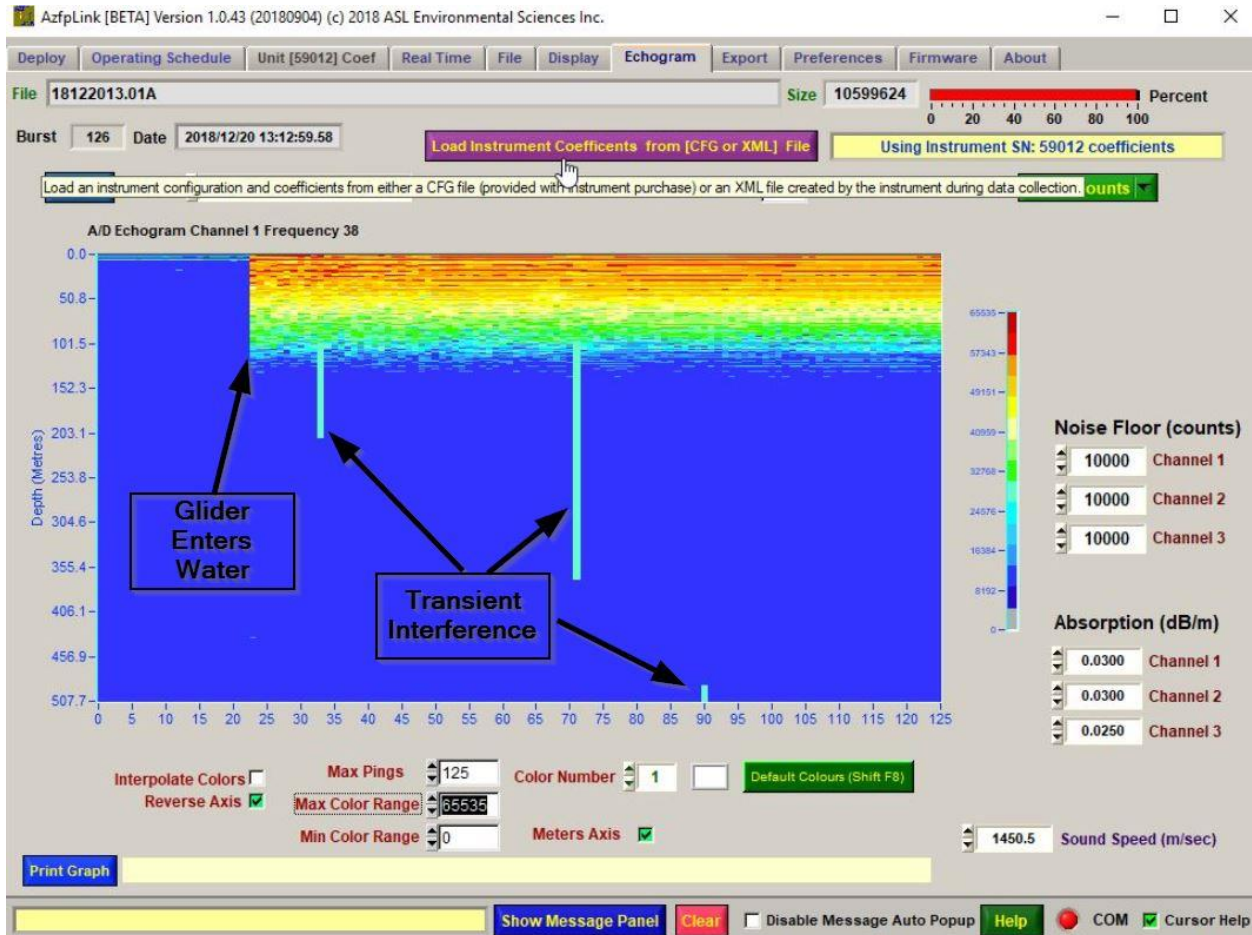


Figure 5: Echogram from the 38 kHz channel for Test 2A of the glider integration tests.

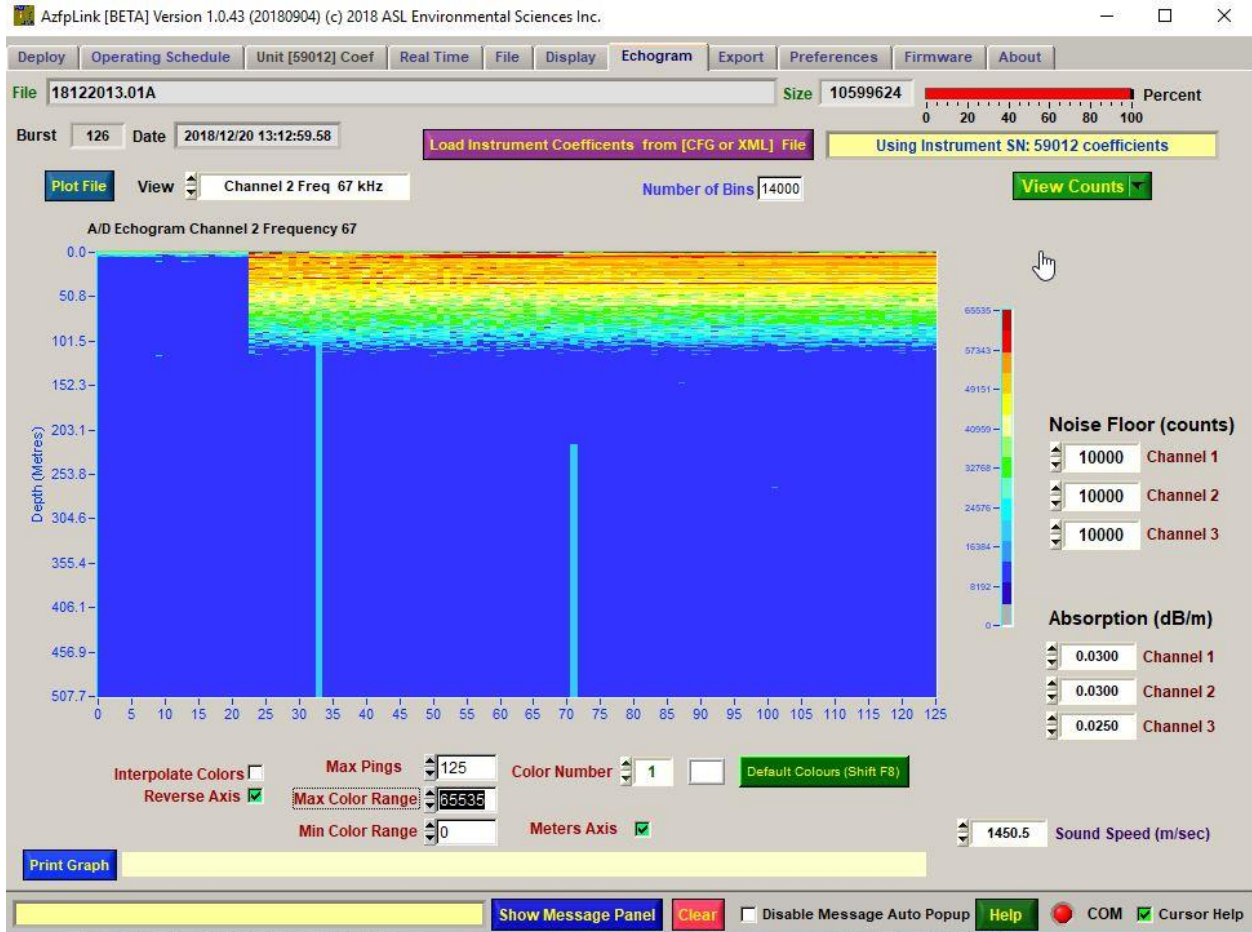


Figure 6: Echogram from the 67 kHz channel for Test 2A of the glider integration tests.

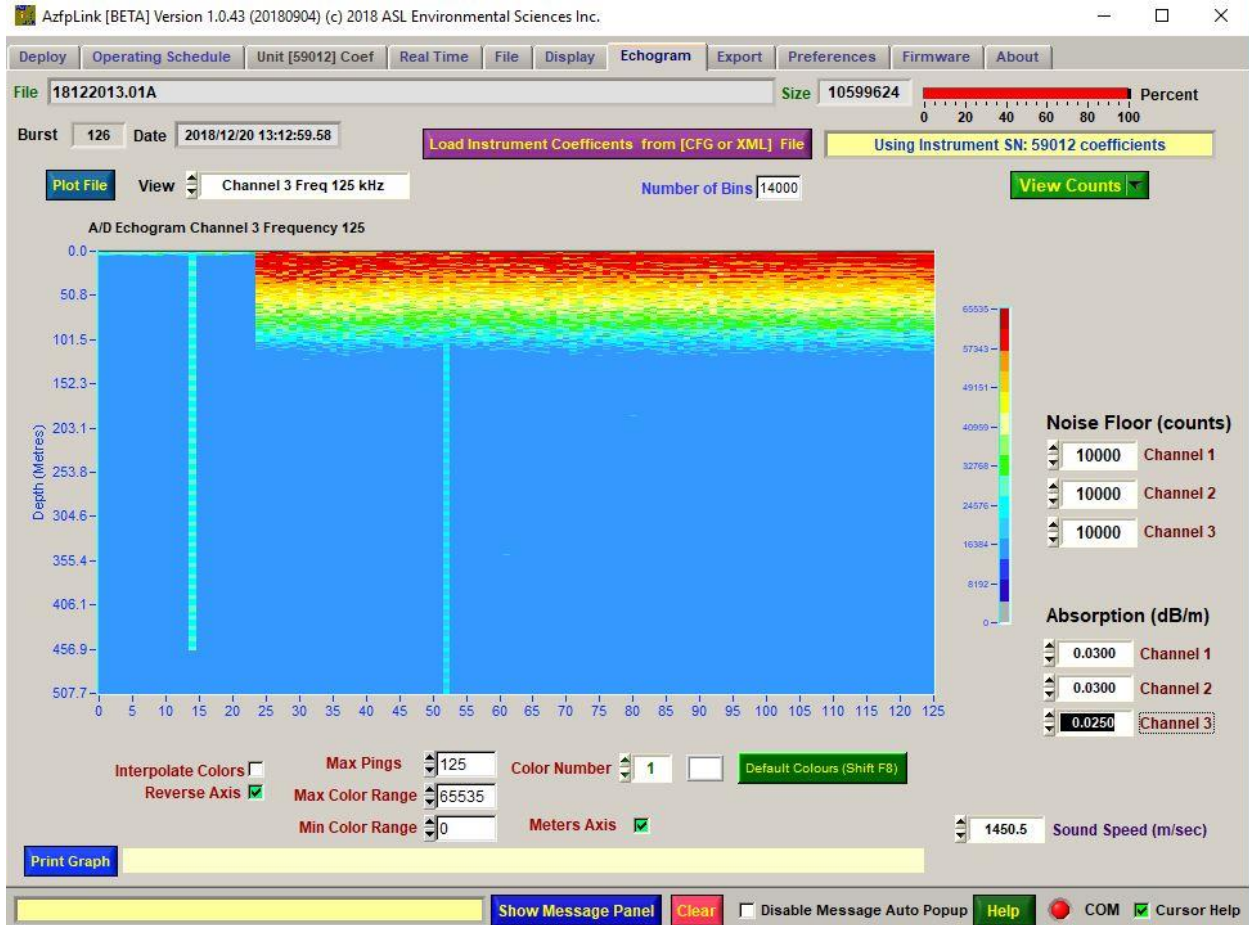


Figure 7: Echogram from the 125 kHz channel for Test 2A of the glider integration tests.

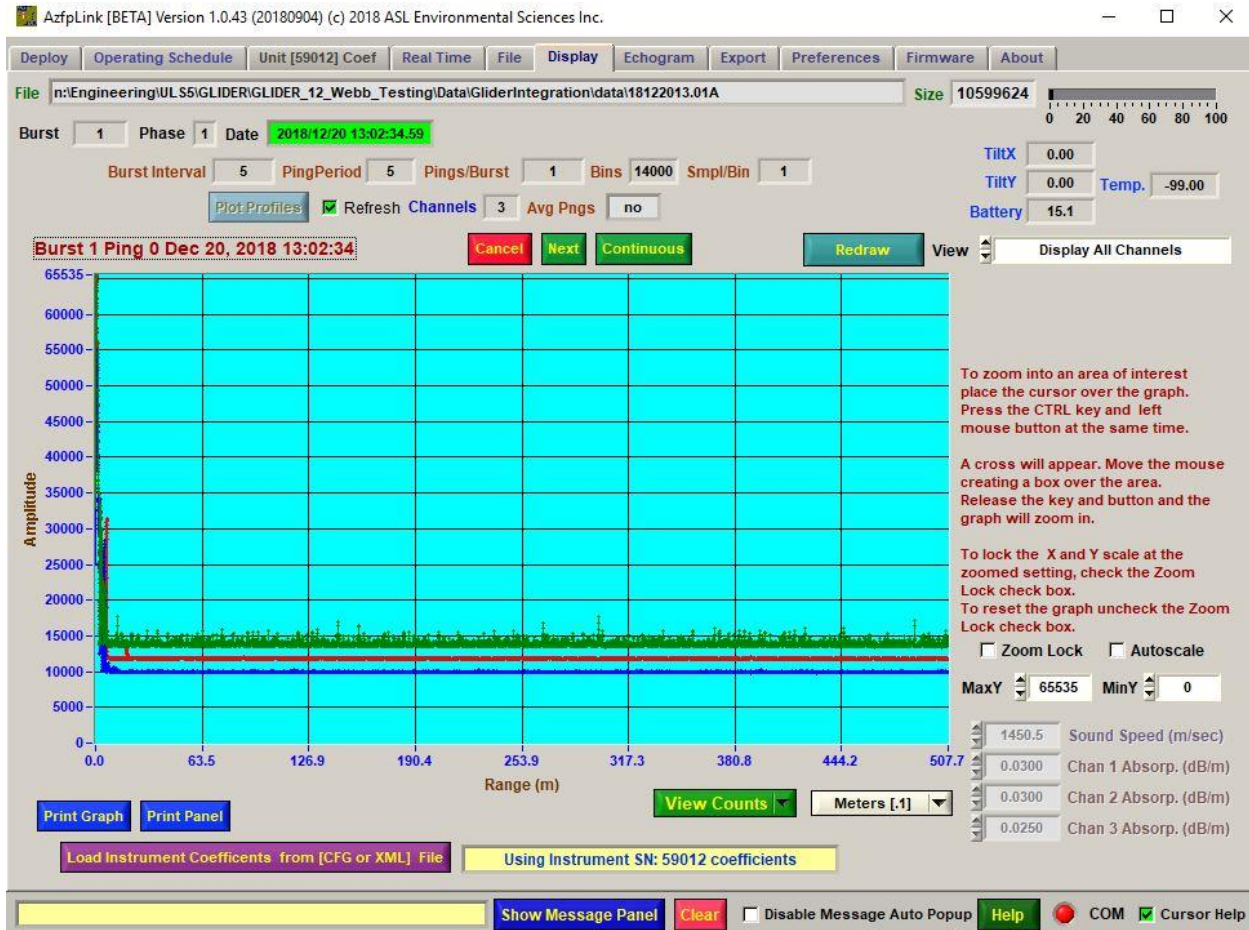


Figure 8: Profile (all channels) for start of Test 2A of the glider integration tests. Noise floors are approximately 10k / 12k / 14k counts on the 67 kHz / 38 kHz / 125 kHz channels, respectively.

Test 2B: Rx Test

The second test of the AZFP module in the fully-assembled glider was an Rx-only test (i.e. the length of the transmit pulse was 0 us). While this test occurred, various sensors and equipment (e.g. pumps and thrusters) were operated. The glider was suspended in the 5x5x5 m tank for this test.

Results once again indicate that the AZFP was running as expected post-integration into the glider. As with Test 2A, noise floors for each channel are again consistent with expectation and with prior testing of the GLIDER-12 module (10k counts on the 67 kHz channel, 12k counts on the 38 kHz channel, and 14k counts on the 125 kHz channel). However, interference is seen on each channel during this test.

The 38 kHz data contain multiple interfering signals, and the echogram is shown in Figure 9 below. One such interfering signal (which is common across all AZFP channels) is the periodic short-duration decaying pulse that is seen at different ranges throughout the data file. This first signal is presumed to be from the Slocum's 172 kHz altimeter, which was operational during this test. The second interfering signal (labeled 'Unknown Interference in Figure 9') appears across four pings. The source of this signal has not been identified. The third and final interfering signal (labeled 'Coincident with thruster activation' in Figure 9) was observed only when the glider's thruster was activated, and did not appear at any other time. It is therefore suspected that this third interfering signal is related to thruster operation.

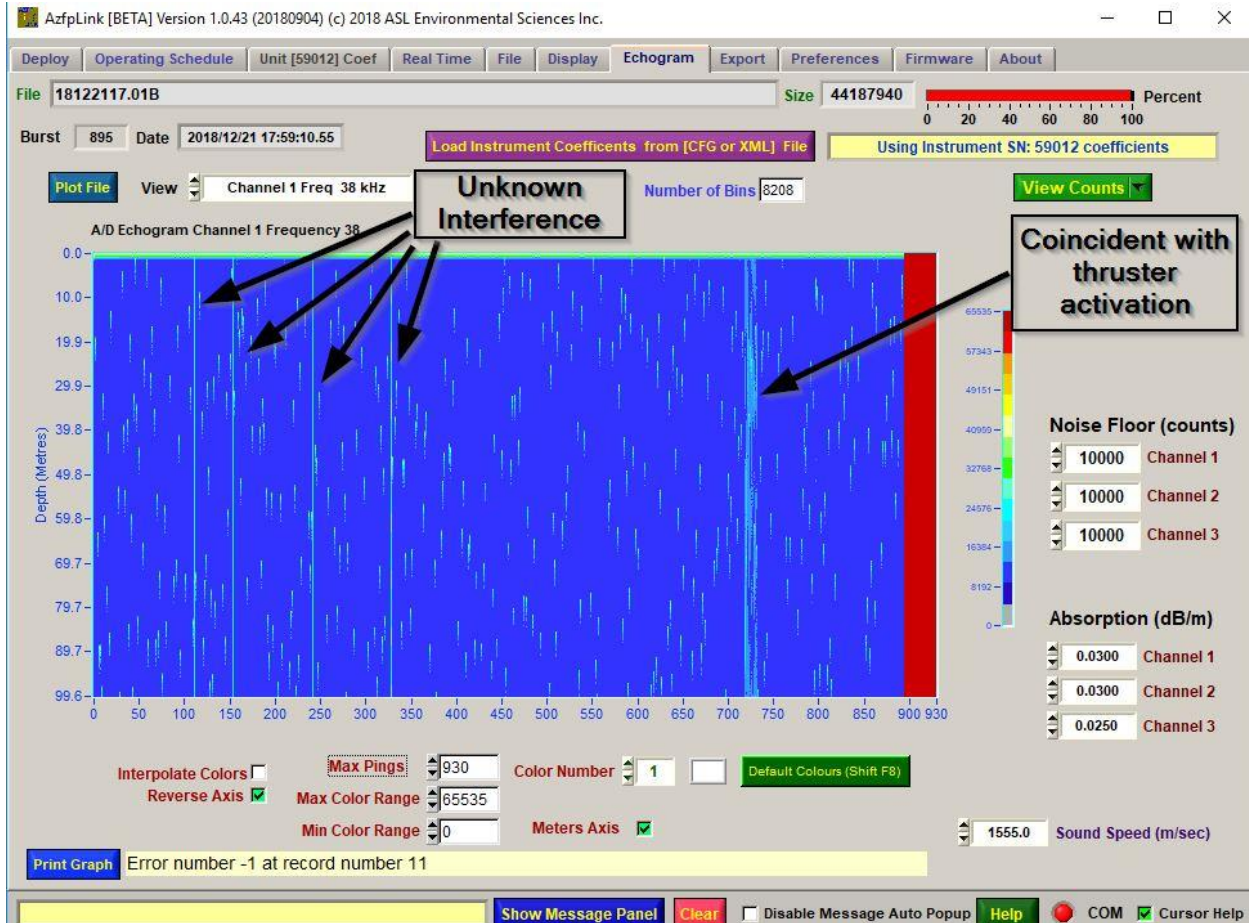


Figure 9: Annotated echogram of the 38 kHz channel for Test 2B of the glider integration tests. Annotations point towards two unique interfering signals, while short-duration interference (presumed to be from the Slocum's 172 kHz altimeter) is seen on almost every ping and different ranges throughout the file.

The 67 kHz channel data contain the short-duration pulses that appear at different ranges in almost every ping (presumed to be interference from the 172 kHz altimeter), and the 'unknown interference' seen in the 38 kHz data. No interference from thruster operation is evident.

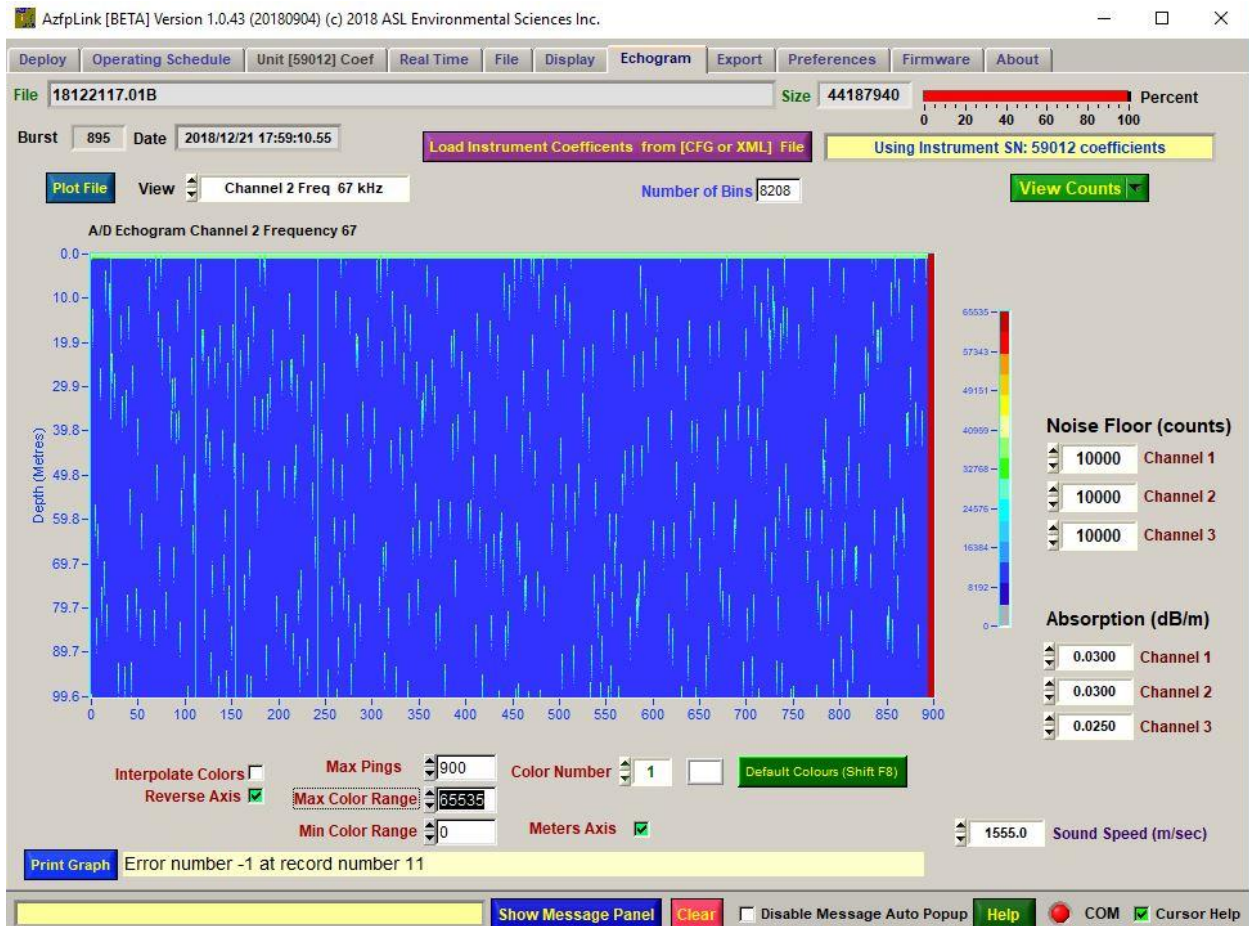


Figure 10: Echogram from the 67 kHz channel during Test 2B of the glider integration tests.

The 125 kHz data show only the interference that is suspected to come from the 172 kHz altimeter; no 'Unknown Interference' or thruster-induced interference is evident.

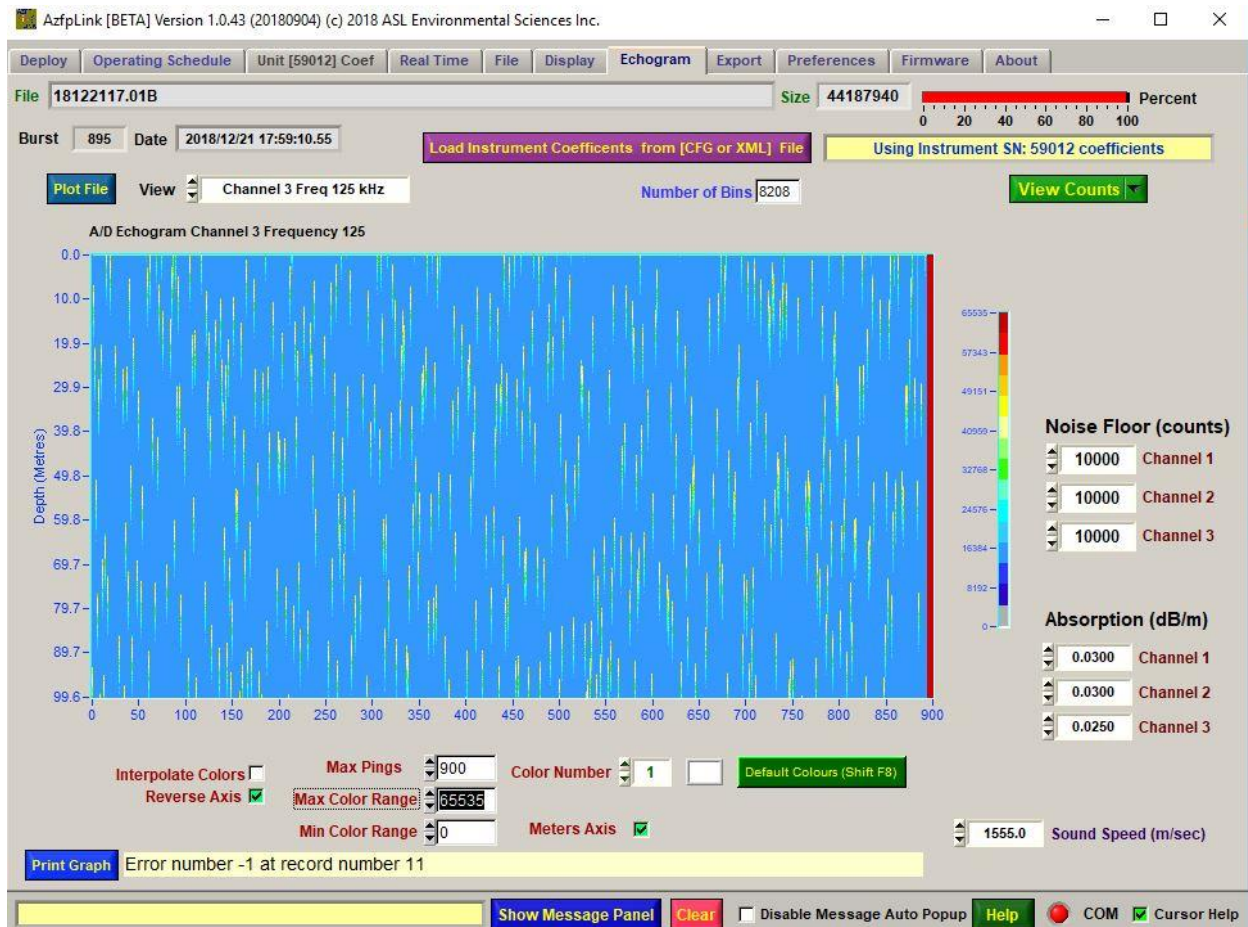


Figure 11: Echogram from the 125 kHz channel for Test 2B of the glider integration tests.

Lastly, a profile (A/D counts vs range for each AZFP channel) is shown for a representative ping. This profile establishes that the noise floors for each channel are consistent with expectations (i.e. ~10k counts on the 67 kHz channel, ~12k counts on the 38 kHz channel, and ~14k counts on the 125 kHz channel) in the absence of the suspected 172 kHz altimeter interference. The 125 kHz channel was the most sensitive to this interference (i.e. the interference had the greatest amplitude and the longest duration).

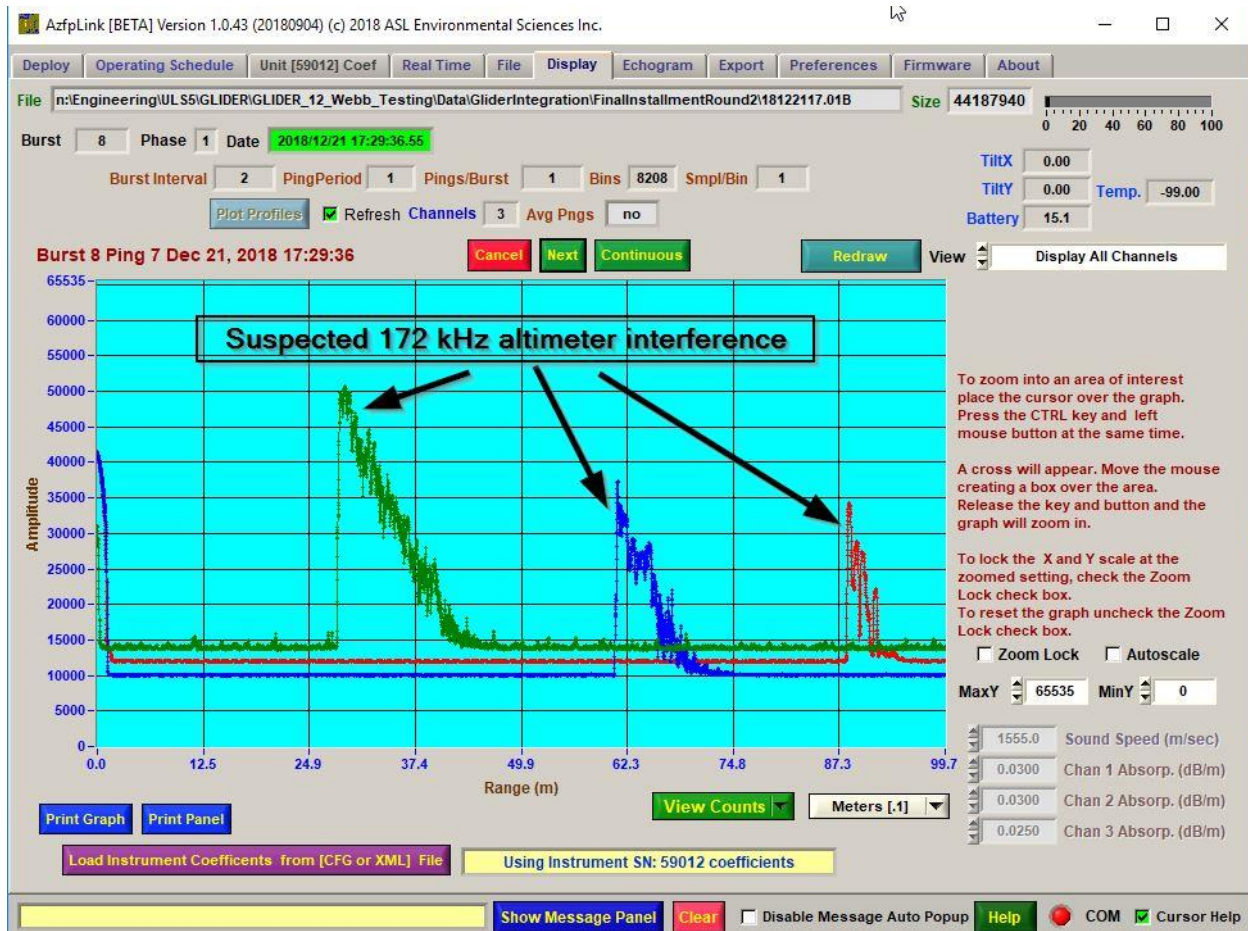


Figure 12: Annotated profile showing the suspected altimeter interference on each channel for Test 2B. In the absence of this interference, noise floors are consistent with earlier measurements.

Conclusions

- The AZFP module 'GLIDER-12' is functioning in a manner that is consistent with expectations;
 - Basic Tx functionality has been verified;
 - Noise level measurements, post-integration into the glider, are consistent with measured noise levels during calibration at ASL.
- Interference has been observed on all channels;
 - The interference was not observed when the AZFP was run as a stand-alone module at ASL or when integrated into the payload bay;
 - Interference was only observed when other sensors and equipment were functioning (e.g. thruster, 172 kHz altimeter);
 - The sources of the interference have not been confirmed but are strongly suspected to include the thruster and the 172 kHz altimeter.
- Interference is believed to be manageable, assuming that the thruster and the altimeter are responsible for this interference;
 - Consider timing the operation of the thruster so that interference with the 38 kHz channel of the AZFP is minimized;
 - Consider timing the operation of the 172 kHz altimeter to minimize interference with the AZFP.

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